SRI RAMACHANDRA
INSTITUTE OF HIGHER EDUCATION AND RESEARCH
(Category - I Deemed to be University) Porur, Chennai
SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

CAD23DLU04: DATA WAREHOUSING & DATA MINING LABORATORY

PROJECT REPORT

SMART CLINICAL NOTES ANALYSIS USING NLP

Submitted by

KOMALA VALLI S - E5223009

BACHELOR OF SCIENCE

IN

ARTIFICIAL INTELLIGENCE AND DATA ANALYTICS

Sri Ramachandra Faculty of Engineering and Technology

Sri Ramachandra Institute of Higher Education and Research,

Porur, Chennai – 600116

OCTOBER, 2025

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BONAFIDE CERTIFICATE

Certified that this project report **"SMART CLINICAL NOTES ANALYSIS USING NLP"** is the Bonafide record of work done by **"KOMALA VALLI S - E5223009".**

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ACKNOWLEDGEMENT

First and foremost, I would like to express my deepest gratitude to **Prof. T. Ragunathan,** Dean, and **Prof. A. Saravanan**, Vice-Dean of Sri Ramachandra Faculty of Engineering and Technology, for their esteemed leadership, encouragement, and for providing an excellent academic environment that enabled the successful completion of this work.

I extend my sincere thanks to **Dr. A. Christopher Tamilmathi,** our Programme Coordinator, for her constant support, valuable advice, and for ensuring the necessary facilities were available to carry out this study.

I am profoundly indebted to my faculty supervisor, **Dr. V Jeevika Tharini**, Department of Computer Science, for her expert guidance, unwavering patience, and relentless encouragement throughout this project. Without her insightful feedback and persistent help, this endeavour would not have reached its successful completion.

My heartfelt appreciation goes to all the faculty members, staff, and my beloved parents and friends for their endless support, motivation, and assistance in overcoming challenges during this journey.

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SMART CLINICAL NOTES ANALYSIS USING NLP

ABSTRACT

Healthcare records contain a large amount of text data in the form of medical transcriptions. These transcriptions include important information about patients such as symptoms, diseases, and conditions. Analyzing this unstructured data can help doctors and researchers find useful patterns for diagnosis and treatment. In this project, we applied data mining and data warehouse techniques to study a medical transcription dataset. The main goal was to extract common symptoms and diseases, measure how often they occur, and discover which ones appear together.

We used the dataset mtsamples.csv and also added some user-provided clinical notes to test our method. A list of common medical keywords such as fever, cough, chest pain, diabetes, infection, nausea, and headache was created. Using Python tools like Pandas, SpaCy, Matplotlib, and Seaborn, we extracted these terms from the notes. The frequency of each symptom was calculated and shown through bar charts and word clouds, which made it easier to see the most common health problems in the dataset.

To study deeper connections, we built a binary matrix of symptoms and applied correlation analysis. This showed strong links like chest pain with shortness of breath and fever with infection. These results prove that combining NLP with data mining can uncover meaningful medical insights from unstructured text. The project highlights how such techniques can support healthcare by making sense of large amounts of patient data, and in the future, more advanced NLP models can be used for greater accuracy.

CHAPTER 1

INTRODUCTION

The healthcare industry generates large amounts of unstructured text data every day through medical notes, prescriptions, and clinical reports. These records contain valuable information about patients' health conditions, symptoms, and treatments. However, because they are written in free-text format, it is difficult to analyze and extract meaningful insights manually. This project uses Natural Language Processing (NLP) and text mining techniques to automatically extract and analyze symptoms and diseases from medical transcriptions, helping convert unstructured text into useful healthcare information.

1.1 OVERVIEW

This project analyzes the Kaggle Medical Transcriptions Dataset (MTSamples), which includes thousands of real-world medical notes. The text is processed using Python tools like Pandas, SpaCy, Matplotlib, and Seaborn to identify and visualize common symptoms and disease patterns. The study focuses on extracting medical keywords, analyzing their frequency, finding correlations, and displaying the results through various charts and word clouds. The system also supports real-time analysis by allowing users to input their own clinical notes.

1.2 PROBLEM STATEMENT

Most medical records are unstructured and stored as free-text, making it hard for healthcare professionals to quickly identify patterns or common symptoms. Manual analysis is time-consuming and error-prone. Hence, there is a need for an automated method that can extract, organize, and visualize important health-related terms from medical transcriptions efficiently.

1.3 SIGNIFICANCE OF THE STUDY

This study is important because it shows how medical data can be used to find useful health patterns. By analyzing medical transcriptions, doctors and researchers can easily identify common symptoms and their relationships. It helps in understanding which health problems are most frequent and which symptoms often appear together. This can support faster diagnosis, better treatment planning, and improved patient care. The study also proves how data mining tools like Python can help in simplifying large medical data into clear and meaningful insights.

1.4 SCOPE OF STUDY

The main scope of this project is to analyze unstructured clinical text data using text mining and NLP to extract and study symptoms and diseases. It focuses on frequency analysis, correlation between symptoms, and visual representation of results. The study demonstrates how unstructured medical notes can be transformed into structured insights that support doctors, researchers, and healthcare organizations in making better clinical and research decisions.

CHAPTER 2

OBJECTIVES

- **To analyze medical transcription data:** The project aims to study real medical transcription data from different medical specialties to understand how symptoms are described in clinical notes.
- **To extract key medical terms and symptoms:** The main goal is to identify and extract common medical keywords such as fever, cough, chest pain, and infection from unstructured text data using Python.
- **To perform frequency analysis:** The study focuses on counting how many times each symptom appears to find the most frequently occurring symptoms in the dataset.
- **To conduct correlation analysis:** The objective is to discover relationships between symptoms, such as how often two symptoms appear together, by creating and analyzing a correlation matrix.
- **To visualize medical patterns effectively:** The project uses different types of visualizations like bar graphs, heatmaps, and word clouds to make the analysis easy to interpret and understand.
- **To support healthcare decision-making:** The final goal is to demonstrate how data analytics can help doctors and hospitals detect health trends, improve diagnosis accuracy, and enhance patient care.

CHAPTER 3

DATASET OVERVIEW

The "Medical Transcription Samples" dataset, sourced from mtsamples.com, contains 4999 text-based medical transcriptions. Its primary purpose is to support Natural Language Processing (NLP) and classification tasks, particularly by medical specialty, Users should be aware that, as "samples". The dataset includes 7 descriptive features per record. Table 3.1.

3.1 TABLE OF DATASET DESCRIPTION

Feature	**Description**
Dataset Name	Medical Transcription Samples
Source	mtsamples.com (scraped data)
Purpose	Provides medical transcription samples to circumvent HIPAA data privacy issues.
Content Type	Text (medical transcriptions)
Key Variable	Medical Specialty (for classification)
Potential Use	Classification tasks, natural language processing (NLP) in healthcare.
Limitations	Data is "samples" and may not represent the full complexity of real-world data.
Acknowledgements	mtsamples.com
Feature Count	7
Record Count	4999

Data Quality

- Missing values:
    - transcription: 33 missing entries
    - keywords: 1,068 missing entries

- Other columns are complete.

Columns Description

1. Unnamed**:** 0 – Row index (not useful for analysis).
2. description – A short summary of the medical case or consultation.
3. medical_specialty – The clinical specialty related to the transcription (e.g., Allergy / Immunology, Bariatrics, Cardiology, Neurology, etc.).
4. sample_name – The case or transcription title (e.g., Allergic Rhinitis, Laparoscopic Gastric Bypass Consult).
5. transcription – The full text of the medical report or consultation note. This is the main body of the dataset.
6. keywords – A list of medical keywords extracted from the transcription (useful for retrieval and topic modeling).

CHAPTER 4

METHODOLOGY

1. **Data Collection:** The Medical Transcriptions Dataset (MTSamples) from Kaggle was used, containing thousands of real medical records. The main field analyzed was the *transcription text*.
2. **Data Preprocessing:** Text data was cleaned, converted to lowercase, and searched for common medical keywords like fever, cough, diabetes, and infection**.**
3. **Symptom Extraction:** A Python function extracted these keywords from each note, adding them to a new column called Extracted_Terms. User notes were also tested for real-time analysis.
4. **Frequency Analysis:** Extracted symptoms were counted to find the most frequent ones, and a word cloud was created for quick visualization.
5. **Correlation Analysis:** A binary matrix and correlation matrix were built to study how symptoms relate to each other. Strong correlations (≥ 0.5) indicated commonly co-occurring symptoms.
6. **Visualization & Interpretation:** Graphs like bar charts, word clouds, and heatmaps helped visualize symptom patterns and relationships, showing how text mining and NLP can turn unstructured healthcare data into useful insights.

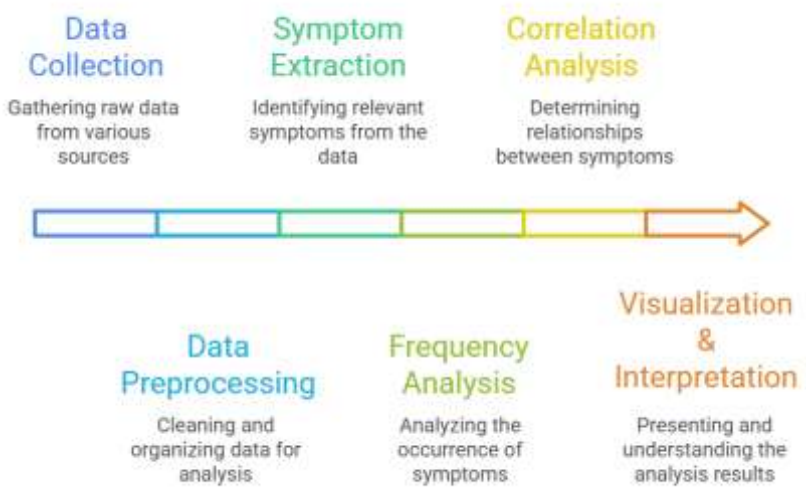

FIG 4.1 METHODOLOGY

CHAPTER 5

RESULT AND DISCUSSION

1.SAMPLE DATASET - The image shows the first five rows of the medical transcription dataset containing fields like “description, medical_specialty, sample_name, transcription, and keywords”. Figure 5.1.

```
Dataset shape: (4999, 6)
   Unnamed: 0                                        description  \
0           0   A 23-year-old white female presents with comp...
1           1           Consult for laparoscopic gastric bypass.
2           2           Consult for laparoscopic gastric bypass.
3           3                              2-D M-Mode. Doppler.
4           4                                 2-D Echocardiogram

             medical_specialty                                sample_name  \
0         Allergy / Immunology                            Allergic Rhinitis
1                   Bariatrics   Laparoscopic Gastric Bypass Consult - 2
2                   Bariatrics   Laparoscopic Gastric Bypass Consult - 1
3   Cardiovascular / Pulmonary                       2-D Echocardiogram - 1
4   Cardiovascular / Pulmonary                       2-D Echocardiogram - 2

                                        transcription  \
0  SUBJECTIVE:,  This 23-year-old white female pr...
1  PAST MEDICAL HISTORY:, He has difficulty climb...
2  HISTORY OF PRESENT ILLNESS: , I have seen ABC ...
3  2-D M-MODE: , ,1.  Left atrial enlargement wit...
4  1.  The left ventricular cavity size and wall ...

                                            keywords
0  allergy / immunology, allergic rhinitis, aller...
1  bariatrics, laparoscopic gastric bypass, weigh...
2  bariatrics, laparoscopic gastric bypass, heart...
3  cardiovascular / pulmonary, 2-d m-mode, dopple...
4  cardiovascular / pulmonary, 2-d, doppler, echo...
```

FIG 5.1 SAMPLE DATASET

2. Clinical Notes with Extracted Symptoms: It shows a table mapping clinical notes to their corresponding extracted symptoms/diseases.It demonstrates how the system successfully identifies key medical terms from unstructured patient notes. Figure 5.2.

	Clinical_Note	Extracted_Terms
0	Patient complains of abdominal pain and vomiting after eating, possible food poisoning	['pain']
1	Reports blurred vision and frequent headaches, possible migraine or eye strain	['headache']
2	Experiencing chest tightness and coughing at night, could be asthma	['cough']
3	Patient has high fever, body ache, and sore throat, suspected viral infection	['fever', 'infection']
4	Complains of burning sensation while urinating, possible urinary tract infection	['infection']
5	Reports back pain and stiffness in the morning, possible muscle strain	['pain']
6	Patient reports sadness, loss of interest, and low energy, possible depression	[]
7	Experiencing pain in the lower abdomen and irregular periods, possible hormonal issue	['pain']
8	Reports shortness of breath and swelling in legs, may indicate heart failure	['shortness of breath']
9	Patient complains of severe headache and confusion, possible brain infection	['headache', 'infection']

FIG 5.2 CLINICAL NOTES WITH EXTRACTED SYMPTOMS

3. Bar chart of the top 10 symptoms and diseases-It highlights pain as the most frequent, followed by infection, fever, chest pain, nausea, and diabetes. Figure 5.3.

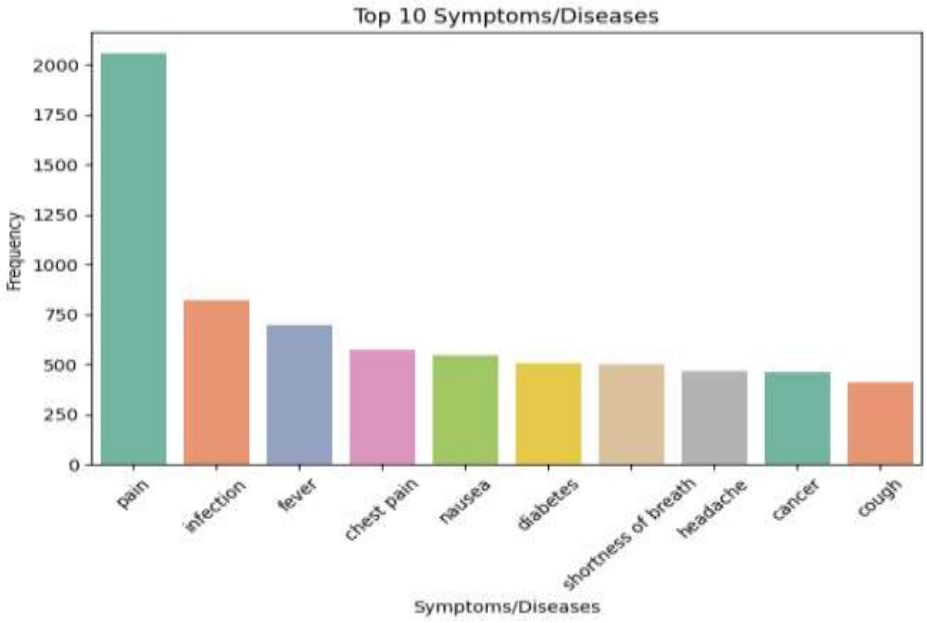

FIG 5.3 BAR CHART OF THE TOP 10 SYMPTOMS AND DISEASES

4.Word Cloud of Symptoms and Diseases-The image shows a word cloud of symptoms and diseases extracted from the dataset.Larger words like pain, chest, infection, and fever indicate their higher frequency in clinical notes. Figure 5.4.

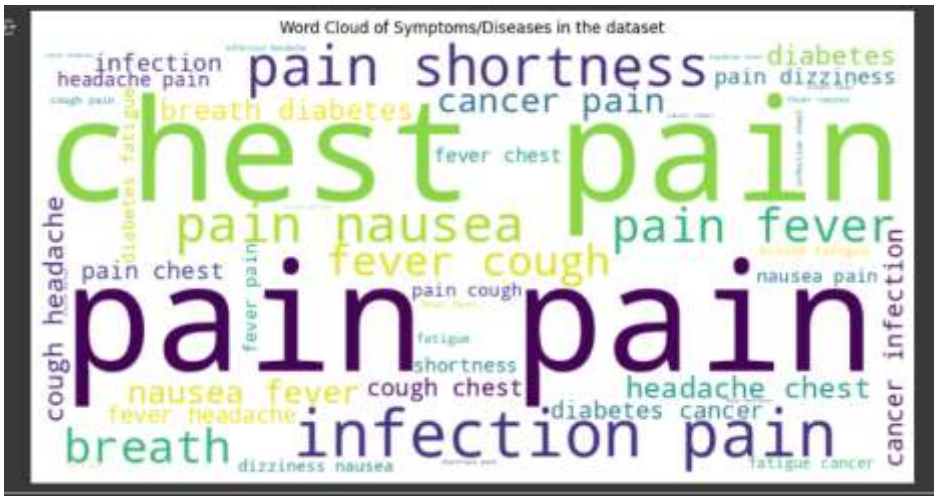

FIG 5.4 WORD CLOUD OF SYMPTOMS AND DISEASES IN DATASET

5.cumulative frequency plot of symptoms and diseases -It reveals that pain, infection, and fever occur much more frequently compared to other symptoms like dizziness and fatigue. Figure 5.5.

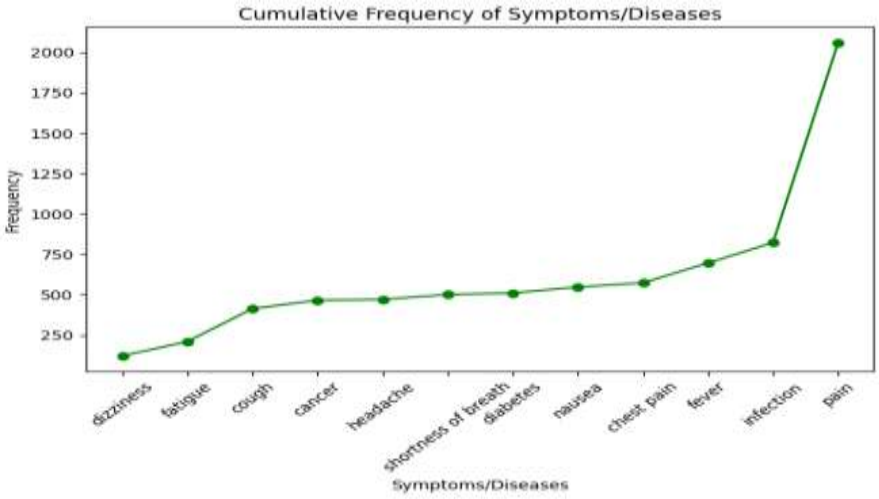

FIG 5.5 CUMULATIVE FREQUENCY PLOT OF SYMPTOMS AND DISEASES

6.correlation matrix of symptoms and diseases -It highlights how certain symptoms co-occur frequently, such as chest pain with shortness of breath and fever with cough. Figure 5.6.

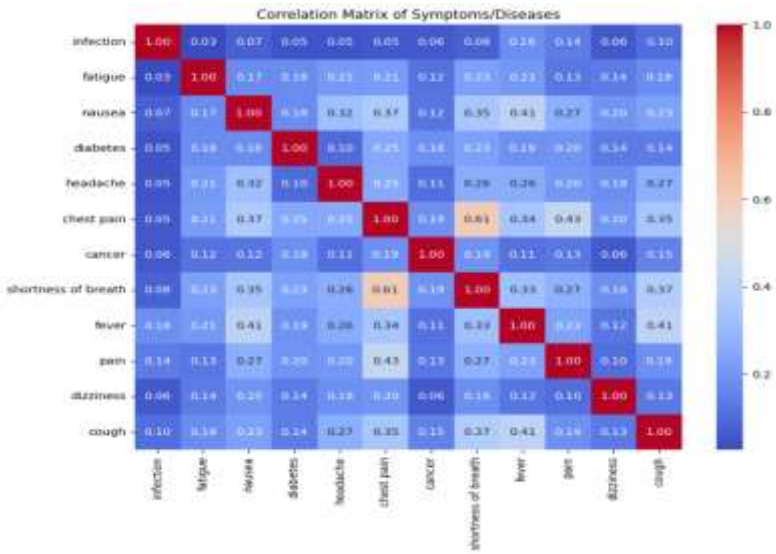

FIG 5.6 CORRELATION MATRIX OF SYMPTOMS AND DISEASES

7.Word Cloud of Symptoms and Diseases-The image shows a word cloud of symptoms and diseases extracted from the user dataset .Larger words like pain, chest, infection, and fever indicate their higher frequency in clinical notes. Figure 5.7.

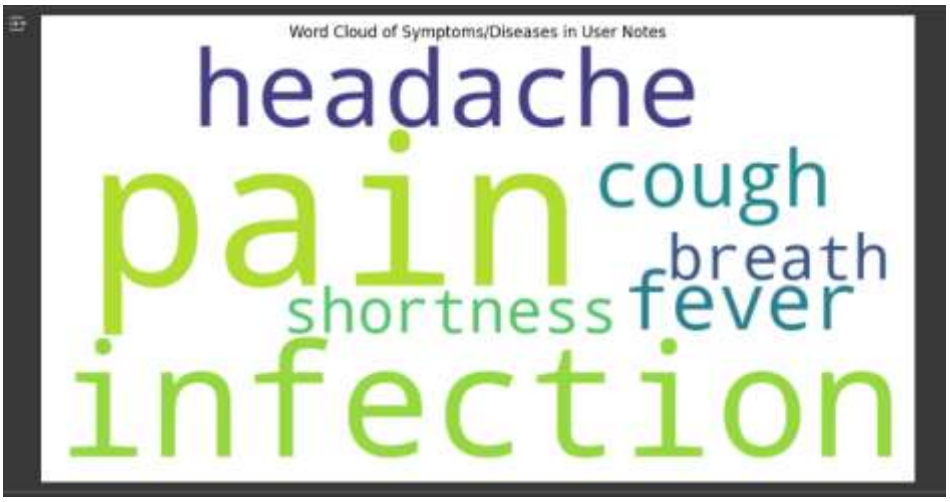

FIG 5.7 WORD CLOUD OF SYMPTOMS AND DISEASES IN USER

CHAPTER 6

CONCLUSION

In this project, we applied data mining and text analysis techniques on medical transcription notes. By using Python and NLP, we were able to extract important symptoms and diseases from unstructured text. We also analyzed how often each symptom occurs and identified patterns in the data.Visualizations such as bar charts, word clouds, and line plots helped in understanding the most common symptoms in the dataset. Correlation analysis showed which symptoms tend to appear together, which can be useful for doctors to detect related health issues.

The project demonstrates that even a simple keyword-based approach can turn large amounts of unstructured medical text into meaningful insights. This method can support healthcare professionals in understanding patient trends and improving decision-making.Overall, the project shows the potential of combining data mining, NLP, and visualization techniques to make medical data more useful and actionable.

6.1 FUTURE SCOPES

- Real-time Sentiment Prediction System
- Explainable AI for Customer Review Analysis
- Advanced Deep Learning for Sentiment Classification
- Product Recommendation Systems Based on Reviews
- AI-powered Customer Support Insights

REFFERENCE

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- Luo, Y., Szolovits, P., & Dligach, D. (2025). Applying text-mining to clinical notes: The identification of patient characteristics. BMC Medical Informatics and Decision Making. **https://bmcmedinformdecismak.biomedcentral.com/articles/10.1186/s12911-025-03137-x**

APPENDICES

APPENDIX-1: CODE – TECHNICAL DETAIL

```
import pandas as pd
import spacy
from collections import Counter
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
from tabulate import tabulate
try:
    df = pd.read_csv('/content/mtsamples.csv', engine='python', on_bad_lines='skip')
    print("Dataset shape:", df.shape)
    print(df.head())
except Exception as e:
    print(f"Error loading dataset: {e}")
nlp = spacy.load("en_core_web_sm")
medical_keywords = ["fever", "cough", "headache", "chest pain",
                    "shortness of breath", "diabetes", "fatigue", "cancer",
                    "infection", "pain", "dizziness", "nausea"]
def extract_terms(note):
    note_lower = str(note).lower()
    terms_found = []
    for word in medical_keywords:
        if word in note_lower:
            terms_found.append(word)
    return terms_found
df['Extracted_Terms'] = df['transcription'].apply(extract_terms)
new_notes = [
"Patient complains of abdominal pain and vomiting after eating, possible food poisoning",
    "Reports blurred vision and frequent headaches, possible migraine or eye strain",
```

```
    "Experiencing chest tightness and coughing at night, could be asthma",
    "Patient has high fever, body ache, and sore throat, suspected viral infection",
    "Complains of burning sensation while urinating, possible urinary tract infection",
    "Reports back pain and stiffness in the morning, possible muscle strain",
    "Patient reports sadness, loss of interest, and low energy, possible depression",
    "Experiencing pain in the lower abdomen and irregular periods, possible hormonal issue",
    "Reports shortness of breath and swelling in legs, may indicate heart failure",
    "Patient complains of severe headache and confusion, possible brain infection"]
new_terms = [extract_terms(note) for note in new_notes]
# Create DataFrame for user notes
user_df = pd.DataFrame({
    "Clinical_Note": new_notes,
    "Extracted_Terms": new_terms})
all_terms = [term for sublist in df['Extracted_Terms'] for term in sublist] + \
            [term for sublist in new_terms for term in sublist]
term_counts = Counter(all_terms)
term_df = pd.DataFrame(term_counts.items(),
columns=['Term', 'Frequency']).sort_values(by='Frequency', ascending=False)
print(tabulate(user_df, headers='keys', tablefmt='grid'))
sns.barplot(x='Term', y='Frequency', data=term_df.head(10), palette='Set2')
plt.title("Top 10 Symptoms/Diseases")
plt.xlabel("Symptoms/Diseases")
plt.ylabel("Frequency")
plt.show()
wc = WordCloud(width=800, height=400, background_color='white')
wc.generate(" ".join(all_terms))
plt.imshow(wc, interpolation='bilinear')
plt.title("Word Cloud of Symptoms/Diseases in the dataset")
plt.show()
sorted_terms = dict(sorted(term_counts.items(), key=lambda item: item[1]))
```

```
plt.plot(list(sorted_terms.keys()), list(sorted_terms.values()), marker='o', color='green')
plt.title("Cumulative Frequency of Symptoms/Diseases")
plt.xlabel("Symptoms/Diseases")
plt.ylabel("Frequency")
plt.show()
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
combined_notes = pd.concat([
    df[['transcription', 'Extracted_Terms']].rename(columns={'transcription':'Clinical_Note'}),
    user_df[['Clinical_Note', 'Extracted_Terms']]
], ignore_index=True)
all_terms_list = list(set([term for sublist in combined_notes['Extracted_Terms'] for term in sublist]))
binary_matrix = pd.DataFrame(0, index=combined_notes.index, columns=all_terms_list)
for i, terms in enumerate(combined_notes['Extracted_Terms']):
    for term in terms:
        binary_matrix.at[i, term] = 1
corr_matrix = binary_matrix.corr()
strong_corr = []
threshold = 0.5
for i in corr_matrix.columns:
    for j in corr_matrix.columns:
        if i != j and corr_matrix.loc[i,j] >= threshold:
            strong_corr.append((i, j, corr_matrix.loc[i,j]))
unique_corr = []
seen = set()
for a,b,c in strong_corr:
    if (b,a) not in seen:
        unique_corr.append((a,b,c))
        seen.add((a,b))
```

```
strong_corr_df = pd.DataFrame(unique_corr, columns=['Symptom 1', 'Symptom 2', 'Correlation'])
print("Strongly Correlated Symptom Pairs (Correlation >= 0.5):")
print(strong_corr_df)
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", cbar=True)
plt.title("Correlation Matrix of Symptoms/Diseases")
plt.show()
user_terms_flat = [term for sublist in new_terms for term in sublist]
wc = WordCloud(width=800, height=400, background_color='white')
wc.generate(" ".join(user_terms_flat))
plt.imshow(wc, interpolation='bilinear')
plt.title("Word Cloud of Symptoms/Diseases in User Notes")
plt.show()
```